Case Study

Case study: Deciding which metadata to show for annotation queues (2023, 2024)

🗂️ Initial state

Working in a company where teams train their own machine learning models, you need annotations. A lot of them.
Therefore, we decided to introduce a **task management** feature to get a flexible representation of **annotation and review workflows** and make them **manageable by end-users.**
For this purpose, we had to understand which information a user of the task management needed to see at first glance.

🤕 Reasons for introducing a Task Management

Manual Effort

- the assignment of users to folders was quite cumbersome
- excel lists were required as a workaround for tracking metadata, e.g. to make sure there was no folder lacking annotators nor annotators creating undesired duplicates

Lack of overview

- at some point, the number of folders and documents was getting overwhelmingly large
- the process for reviewing annotations was based on forwarding links back and forth

💡 Steps for developing the concept

Step	Technique	Tooling
Designing a user workflow	Flow Chart, User Journey	Confluence, Figma
Deciding which views are needed	Notes	Confluence
Deciding on details per view • Which information do users need per queue? • Which interactions do users need per queue?	Wireframes, Doodles, Questionnaire	Figma, Browser DevTools, Slack/Polly
Detailed visual mock-up per view	Mockup	Figma

This is the step we'll look into

💡 Designing the task queue overview: Questionnaire

Task queues consist of multiple tasks grouped together. Each queue has **annotators** and **reviewers** assigned to it, as well as different resources to be annotated. When an annotation queue is marked as ready, the system automatically creates a review task queue for evaluation of the annotation tasks.

Since showing all of the available queue metadata would clutter the UI, we decided to involve the users, since, after all, they were the ones who knew how they worked and which information would make their workflow most efficient.

The future users were provided a questionnaire with two multiple-choice questions:

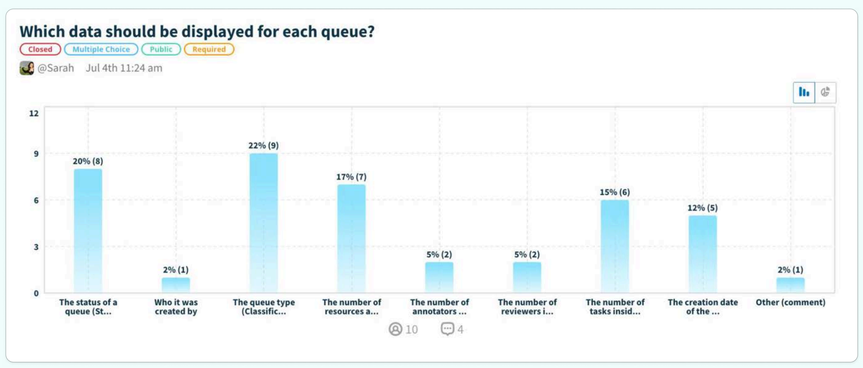
1. **Which data should be displayed for each queue?**
2. **Do you wish to see details on the queue progress?**

Additionally, users had the chance to provide open comments as feedback.

💡💡 Data to show per queue

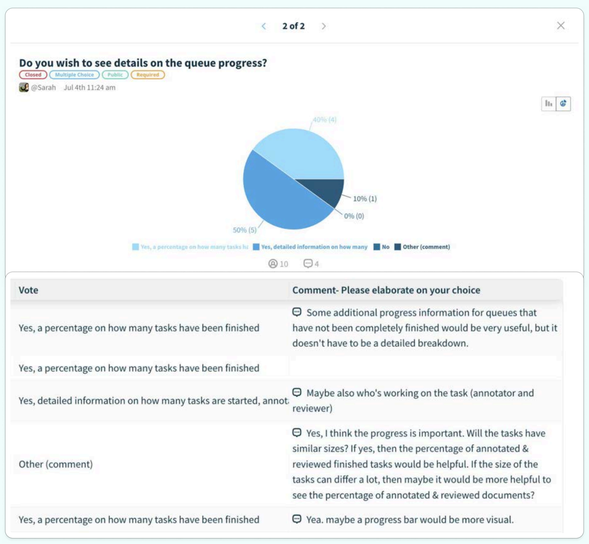
In total, 10 people answered, out of which 7 were part of the future user group.
The top information they desired to see was:

1. **22% Queue Type:** *the type of the annotation task*
2. **20% Queue Status:** *open, started, finished*
3. **17% Number of resources:** *how many documents are in the queue*
4. **15% Number of tasks inside the queue**
5. **12% Creation date of the queue**

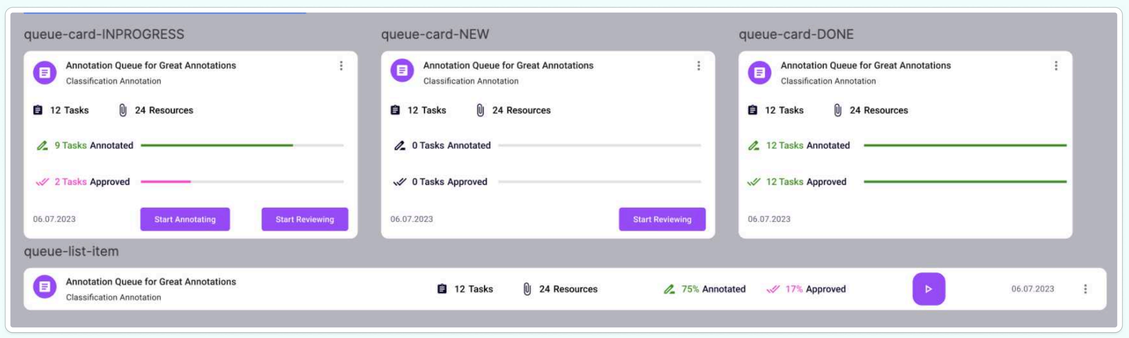


💡💡 Displaying queue progress

As for the queue progress, 100% agreed that it was information they needed to see.
People suggested visual elements, such as a progress bar. They also mentioned interest in viewing information about review progress in addition to the annotation progress.



Next steps & Learnings

- out of the gathered information, we then developed first designs for how such a task queue could be visualized & implemented a view that is still used to date
- users were very grateful for being involved in the overall process
- gathering their feedback via a questionnaire was an efficient way of quickly gathering valuable input with respect to their requirements

- 💡**Thinking more holistically in workflows rather than asking for individual information could have been beneficial**: we later realized that the number of annotators/reviewers was indeed information they needed, despite being rated low in the survey
- 💡 **Survey results are no everlasting design decision:** user flows and user needs are dynamic and can change – frequent check-ins are still necessary
- 💡 **User involvement is no one-time event, but a repeatable good practice.**